We define a policy to be a stochastic mapping $\pi : (\mathcal{A} \times \mathcal{E})^* \to \Delta\mathcal{A}$ from histories to actions, as usual. We define an environment μ to be a pair (T_μ, u_μ). The first component is the transition function $T_\mu : (\mathcal{A} \times \mathcal{E})^* \times \mathcal{A} \to \Delta\mathcal{E}$, and the second component is the utility function $u_\mu : (\mathcal{A} \times \mathcal{E})^\infty \to [0, 1]$ that scores full trajectories. This allows us to model situations where the reward is not observed during interaction.

The value of a policy π, in an environment μ, is defined as the expected utility:

$$V_\mu^\pi = \mathbb{E}_{h \sim \mu^\pi} \left[u_\mu(h) \right].$$

A mixture environment $\xi = \sum_\nu w(\nu)\nu$ is defined with the transition function $T_\xi(e_t \mid h_{<t}a_t) = \sum_\nu w(\nu \mid h_{<t}) T_\nu(e_t \mid h_{<t}a_t)$ and the utility function $u_\xi(h) = \sum_\nu w(\nu \mid h) u_\nu(h)$. We will abuse the notation and write μ for T_μ when there is no ambiguity.

Environment-policy process. Suppose we have an environment generator $\mathrm{M} : (\mathcal{P} \times \mathcal{M})^* \to \Delta\mathcal{M}$ and a policy generator $\Pi : (\mathcal{P} \times \mathcal{M})^* \to \Delta\mathcal{P}$, where $\mathcal{P}$ is a countable class of policies and $\mathcal{M}$ a countable class of environments. Together, they induce the environment-policy process M^Π, defined by

$$\mathrm{M}^\Pi(\pi\mu_{1:n}) = \prod_{i=1}^{n} \Pi(\pi_i \mid \pi\mu_{<i}) \mathrm{M}(\mu_i \mid \pi\mu_{<i}).$$

where $\pi\mu_i$ is shorthand for (π_i, μ_i). Note that Π and M generate π_i and μ_i simultaneously, instead of one after another. This is akin to a multi-agent system with two agents M and Π.

Note:

- I started with a much simpler formulation, where μ_i only depends on $\mu_{<i}$. But then I realized that this is not very realistic, because engineers can be informed by the trained policies, to choose which task the policies should be trained on next. The current formulation doesn't seem very realistic either because μ_i should be able to depend on π_i as well. The reason why I chose the current formulation is because the notion of asymptotic optimality becomes ambiguous and messy with the more realistic formulation.
- It's possible that the generality of the current formulation doesn't get us anywhere. In that case, we should go back to the simpler formualation where μ_i only depends on $\mu_{<i}$.
- I've been side-tracked by the idea of deceptive alignment (alignment faking), and that's why I started thinking about the more general formulation. But the more I think about it, it's not so obvious whether it is feasible to say anything interesting about deceptive alignment with our overall approach. After all, Hubinger et al. believe that the deceptively aligned policies can emerge due to inductive bias, but likely not due to an external incentive.

Standard Policy Optimization

Given a sequence of environments $\mu_{<n}$, a standard "policy optimization" procedure has the objective function

$$J(\pi) = \frac{1}{n-1} \sum_{i=1}^{n-1} V_{\mu_i}^\pi.$$

We can consider the Gibbs posterior

$$\omega_n(\pi) \propto \omega(\pi) \exp\left(\beta_n J(\pi)\right),$$

where $\omega(\pi)$ is a Gibbs *prior* that encodes some inductive bias.

Then, standard policy optimization Π is defined as:

$$\Pi(\pi \mid \pi\mu_{<n}) = \omega_n(\pi).$$

Notes:

- This is a model of perfect optimization, unlike in practice where we have sloppy optimization due to limited compute and data.
- The objective function can be a sum of RL objectives and supervised objectives, e.g., cross-entropy loss. We focus on the RL objective here, but I expect most of the analysis to generalize to the supervised objective.

Asymptotic Optimality

What does it mean for a policy generator Π to be *effective*? This is related to the question of which policy generators we can expect people to use in practice, and somewhat related to how interpretable/predictable the behavior of Π is.

Let $\xi_n = \mathbb{E}_{\mu_n \sim \mathrm{M}(\cdot \mid \pi\mu_{<n})}\left[\mu_n\right]$ be the mixture of environments output by M at time n. Similarly, let $\zeta_n = \mathbb{E}_{\pi_n \sim \Pi(\cdot \mid \pi\mu_{<n})}\left[\pi_n\right]$. We define the asymptotic optimality of Π when coupled with M as follows:

$$\lim_{n\to\infty} V^*_{\xi_n} - V^{\zeta_n}_{\xi_n} = 0,$$

with M^Π-probability 1.

This is equivalent to saying that Π converges to the best response against M, with respect to the *myopic* objective $V^{\pi_n}_{\mu_n}$. We can also imagine a non-myopic objective, in which case an effective Π would generate π_n that steers M toward generating more favorable environments in the future, e.g., environments that are more generous with their rewards. But not only could this be disastrous from the safety perspective, it's also not very useful (or at least I don't see how it could be useful).

IID Environments

Suppose M generates i.i.d. environments, that is,

$$\mathrm{M}(\mu_i \mid \pi\mu_{<i}\pi_i) = \mathrm{M}(\mu_i)$$

for all i. Also, let Π be the standard policy optimization, as described above.

As $n \to \infty$, $\hat{\xi}_n = \frac{1}{n-1}(\mu_1 + \cdots + \mu_{n-1})$ converges to $\xi = \mathbb{E}_{\nu\sim\mathrm{M}}\left[\nu\right]$. Since $J(\pi) = V^\pi_{\hat{\xi}_n}$, the Gibbs posterior ω_n concentrates on optimal policies in ξ. Thus, $\lim_{n\to\infty} V^*_\xi - V^{\zeta_n}_\xi = 0$ holds M^Π-almost surely, and Π is asymptotically optimal. A formal proof with proper conditions on β_n can be found in **Appendix A.1**.

General Environment Process

In general, standard policy optimization is not asymptotically optimal when M is not i.i.d.

Consider, for instance, a simple environment generator M that outputs μ_1 at odd times and μ_2 at even times.

$$\mu_1 \to \mu_2 \to \mu_1 \to \dots$$

Standard policy optimization will converge to a policy that is optimal in the mixture environment $\xi = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2$, which is not necessarily optimal in either μ_1 or μ_2.

Distinguishability

We said an optimal policy in $\xi = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2$ is not necessarily optimal in μ_1 or μ_2. But can it be optimal? What if μ_1 and μ_2 are completely distinguishable from the start, e.g., because they output some rich contexts c_1 and c_2 at the beginning? Then,

$$V^\pi_\xi = \frac{1}{2}V^\pi_{\mu_1} + \frac{1}{2}V^\pi_{\mu_2} = \frac{1}{2}V^{\pi(\cdot|c_1)}_{\mu_1}(c_1) + \frac{1}{2}V^{\pi(\cdot|c_2)}_{\mu_2}(c_2),$$

and since $\pi(\cdot \mid c_1)$ and $\pi(\cdot \mid c_2)$ are independent,

$$V^*_\xi = \frac{1}{2}V^*_{\mu_1} + \frac{1}{2}V^*_{\mu_2}.$$

Similarly, suppose M generates environments ν that output a unique context c_ν at the beginning. Let $\hat{f}_n(\nu)$ denote the empirical frequency of ν until time $n-1$. If eventually $\mathrm{M}(\nu \mid \pi\nu_{<n}) \le C\hat{f}_n(\nu)$ for all ν, then

$$V^*_{\xi_n} - V^{\zeta_n}_{\xi_n} = \sum_\nu \mathrm{M}(\nu \mid \pi\nu_{<n})\big(V^*_\nu - V^{\zeta_n}_\nu\big) \le C\sum_\nu \hat{f}_n(\nu)\big(V^*_\nu - V^{\zeta_n}_\nu\big) \to 0$$

so Π is asymptotically optimal.

Model-based policy optimization

We showed above that standard policy optimization isn't asymptotically optimal in the general case. Is there a policy generator that is asymptotically optimal in all environment generators M from some class $\mathfrak{M}$? The obvious solution would be to guess the next environment μ_n based on the history $\pi\mu_{<n}$, and then optimize the policy π_n for the predicted environment.

If we have some prior $\mathfrak{w}$ over the class $\mathfrak{M}$ of environment generators, we can perform Bayesian inference to obtain the posterior $\mathfrak{w}_n$ given the history $\pi\mu_{<n}$. Then, we perform standard policy optimization on the mixture environment $\hat{\xi}_n = \mathbb{E}_{\mathrm{M}\sim\mathfrak{w}_n,\, \mu\sim\mathrm{M}(\cdot|\pi\mu_{<n})}[\mu]$. Since $\hat{\xi}_n - \xi_n \to 0$ by merging of opinions, we can probably show that the resulting policy generator Π is asymptotically optimal, under some conditions on β_n.

Learning to do model-based policy optimization

It's possible that under some extra conditions, providing $\pi\mu_{<n}$ as a context to π_n and running standard policy optimization is asymptotically optimal. This is because ξ_n can eventually be inferred from $\pi\mu_{<n}$ with arbitrary precision, and thus π_n can learn to infer ξ_n from $\pi\mu_{<n}$ and optimize for it — essentially learning to do the model-based policy optimization within itself.

Mesa-optimizers

Note:

- This section is pretty unrelated to the previous section on asymptotic optimality. In fact, we only consider a single training environment μ.

How does a policy, obtained through policy optimization on some environment μ, behave in a state never encountered during training on μ? For instance, μ is a maze environment where the goal is to reach a red box. How will the policy trained on μ behave in μ', where the maze has red circles instead? Would it still act in a goal-directed manner? If so, can the goal be predicted from the goal in μ?

Suppose ω is the simplicity prior $\omega(\pi) = \omega(\pi) \propto 2^{-K(\pi)}$, which assigns higher probability to simpler policies in the class $\mathcal{P}$. Let's also define the simplicity prior $w(\nu) \propto 2^{-K(\nu)}$ on environments in class $\mathcal{M}$.

Let $\pi^\dagger_- : \mathcal{M} \to \mathcal{P}$ be a mapping from environments to policies, such that $\pi^\dagger_\nu$ is near-optimal in ν for all $\nu \in \mathcal{M}$,

$$V_\nu^{\pi^\dagger_\nu} \geq V_\nu^* - \varepsilon,$$

and $\pi^\dagger_\nu$ is as about simple as ν, in the sense that

$$-\log \omega(\pi^\dagger_\nu) \overset{+}{=} -\log w(\nu).$$

Suppose also that every near-optimal policy in μ can be represented as $\pi^\dagger_\nu$ for some environment ν. (This is a very strong assumption.)

Let $\mathcal{M}_\mu$ be the set of all environments ν such that a policy being near-optimal in ν implies the same policy being near-optimal in μ. In other words,

$$\mathcal{M}_\mu = \{\nu \in \mathcal{M} : \forall \pi \in \mathcal{P},\ V_\nu^\pi \geq V_\nu^* - \varepsilon \implies V_\mu^\pi \geq V_\mu^* - \varepsilon\}.$$

I want to show that for large β, the Gibbs posterior $\omega_\beta(\pi) \propto \omega(\pi) \exp\left(\beta V_\mu^\pi\right)$ is approximately the pushforward of the distribution $w_\mu(\nu) \propto w(\nu)\mathbb{I}[\nu \in \mathcal{M}_\mu]$ under the mapping $\pi^\dagger_-$.

Appendix

A.1. Proof of asymptotic optimality in the i.i.d. case

Choose β_n such that

$$\log n \ll \beta_n \ll n.$$

For example, $\beta_n = \sqrt{n}$ works. Assume that there exists some policy $\pi^* \in \mathcal{P}$ that is optimal in ξ. (We can relax this assumption to a sequence of policies π_n^* that are ε_n-optimal, with $\varepsilon_n \searrow 0$.)

For a distribution q over $\mathcal{P}$, write $V_\nu^q = \sum_{\pi \in \mathcal{P}} q(\pi) V_\nu^\pi$. The Gibbs posterior ω_n has the variational characterization

$$\omega_n = \arg\max_q V_{\hat{\xi}_n}^q - \frac{1}{\beta_n} \mathrm{KL}(q \| \omega).$$

By two-sided Hoeffding-style PAC-Bayes bound (that follows from Theorem 2.1 in "User-friendly Introduction to PAC-Bayes Bounds"), with probability at least $1 - \delta$, for all posteriors q,

$$|V_\xi^q - V_{\hat{\xi}_n}^q| \le \frac{\mathrm{KL}(q\|\omega) + \log(2/\delta)}{\beta_n} + \frac{\beta_n}{8(n-1)}.$$

Combined with the variational characterization of ω_n, for any distribution q,

$$V_\xi^q - V_\xi^{\omega_n} \le 2\frac{\mathrm{KL}(q\|\omega) + \log(2/\delta)}{\beta_n} + \frac{\beta_n}{4(n-1)}.$$

Taking $q = \delta_{\pi^*}$ gives

$$V_\xi^* - V_\xi^{\omega_n} \le 2\frac{\log\frac{1}{\omega(\pi^*)} + \log(2/\delta)}{\beta_n} + \frac{\beta_n}{4(n-1)}.$$

Now set $\delta_n = n^{-2}$. By Borel-Cantelli, the bound holds eventually almost surely. Since $\log(2/\delta_n) = O(\log n)$, $\log n \ll \beta_n$, and $\beta_n \ll n$, the right-hand side goes to 0. Therefore

$$\lim_{n\to\infty} V_\xi^* - V_\xi^{\omega_n} \to 0$$

almost surely.